

Introduction to Computational Cognitive Modeling

Jussi P. P. Jokinen

Cognitive Science
University of Jyväskylä
Jyväskylä, Finland
jussi.p.p.jokinen@jyu.fi

Antti Oulasvirta

Aalto University
Helsinki, Finland
antti.oulasvirta@aalto.fi

Andrew Howes

Computer Science
University of Exeter
Exeter, United Kingdom
howesamail@gmail.com

Abstract

This course introduces computational cognitive modeling for researchers and practitioners in the field of HCI. Cognitive models use computer programs to model how users perceive, think, and act in human–computer interaction. They offer a powerful approach for understanding interactive tasks and improving user interfaces. This course starts with a review of classic approaches to cognitive modeling. It then rapidly progresses to introducing computational rationality, which uses modern modeling approaches powered by machine learning methods, in particular deep reinforcement learning. The course is built around hands-on Python programming using notebooks.

CCS Concepts

• **Human-centered computing** → *HCI theory, concepts and models*.

Keywords

Cognitive modeling, cognitive architectures, reinforcement learning, computational rationality, cooperative intelligence, user interface optimization

ACM Reference Format:

Jussi P. P. Jokinen, Antti Oulasvirta, and Andrew Howes. 2025. Introduction to Computational Cognitive Modeling. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3706599.3706648>

1 Introduction

Benefits. The course will cover the state-of-the-art in cognitive computational modeling for HCI. The attendees will learn how contemporary modeling paradigms have evolved from the classic approaches used in HCI. They will also get to understand how recent advances in machine learning and cognitive science have made it possible for the state-of-the-art to emerge. After the course, the attendees will be able to approach theoretical and practical HCI problems from the perspective of computational cognitive modeling. They are provided skills and concrete tools with which to apply models in their work, alongside the theoretical basis for how to develop these models further.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI EA '25, Yokohama, Japan

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3706648>

Computational cognitive modeling has re-emerged – after somewhat of a hiatus – as an important and rapidly advancing paradigm for understanding interaction. Integration of recent advances in machine learning and cognitive science thinking has resulted in models that are able to shed light on such key topics of HCI as multitasking [3], gaze-based selection [1], and touchscreen typing [2, 5]. As these models become more advanced and readily employable in HCI research and design, it is important for everyone – not just the modelers – to understand their theoretical and technical background. The motivation for taking this course is to stay up-to-date with the recent developments and learn to utilize them in one's own work, both theoretical and practical.

Intended audience. The course is primarily targeted at researchers and practitioners in HCI with interest in applications of computational methods and machine learning to the problem of understanding users. Industrial practitioners, interested for example in user modeling, may also find the course valuable.

Prerequisites. While the course will cover topics in machine learning, such as deep reinforcement learning, the audience is not expected to have a machine learning background. The course will provide a practical introduction to these topics, and Python notebooks will be used to interactively learn to use existing models and provide ground for the attendees to learn to develop new models. The audience is therefore expected to have Python notebooks installed on their computers, and at least an introductory level knowledge on Python or programming in general.

2 Contents

The course will focus on cognitive models driven by modern ML methods and on applying these techniques to concrete HCI problems.

The course consists of 3 x 75 minute sessions:

- Introduction to computational cognitive modeling
- Computational rationality
- Cognitive modeling for HCI

Each is expanded in the following subsections:

2.1 Introduction to computational cognitive modeling

In this session we will critically review the history of cognitive modeling in HCI over the past 50 years. The review will cover modeling approaches that include mathematical models of movement time, procedural models of skill, models of eye movements, computational simulations of human information processing, models inspired by how animals forage for food and models of human learning, rationality and decision making.

The strengths and weaknesses of each approach will be considered with respect to their contribution, both real and perceived, to HCI. The section discusses the uses of models in HCI, such as advancing scientific understanding, evaluating designs, and facilitating cooperative AI. After completing the section the attendees will understand the role of computational cognitive modeling in HCI, and know how the historical approaches have converged into the state-of-the-art of modeling.

2.2 Computational rationality

In this unit, we cover how to create computational rational models of interaction [4]. The session provides an introduction to creating deep reinforcement learning based simulation models of interactive tasks. We cover the method of defining goals, bounds, and the interaction environment, and how to simulate such task environments to make realistic predictions of complex interaction. After completing this section, attendees understand what the principles of computational rational models of interaction are, and how to derive bounded optimal interactive policies using deep RL.

2.3 Building a model for HCI

In the final unit, we apply what was learned in the previous units and design and build a model of interactive task behavior from starting principles of computational rationality. This module covers the steps required for making computational rational agents, starting from defining the internal and external task environments and the agent's goals. We also cover how to train deep RL agents and how to inspect their predictions. The focus is on investigating how the design of the task environment and the internal cognitive processes interact in a way that is consistent with empirically observed human-like behavior.

2.4 Practical work

The course makes heavy use of Python notebooks. Each session includes a notebook that has been prepared for that session, with activities throughout the session carried out on the notebooks. The notebook platform provides an easy way for the attendees to follow the session and explore and experiment on relevant model code. Each session includes short assignments that have the attendees carry out tasks using the notebooks. The goal of this interactivity is that the attendees get hands-on experience with computational cognitive models, not just theoretical overview and examples.

3 Instructor background

Jussi P.P. Jokinen is Associate Professor of cognitive science at University of Jyväskylä. He was previously a Research Fellow at Finnish Center of AI FCAI. His doctorate is from cognitive science in University of Jyväskylä in 2015. His award-winning work on computational cognitive modeling has been published in multiple HCI and cognitive science venues, such as ACM CHI, IHCS, Human Factors, Cognitive Science.

Antti Oulasvirta is Professor at Aalto University where he leads the Computational Behavior Lab and the Interactive AI research programme at the Finnish Center for AI. He was previously a Senior Researcher at the Max Planck Institute for Informatics. He received his doctorate in Cognitive Science from the University of Helsinki

in 2006. He was awarded an ERC Starting Grant (2015-2020) for research on computational design of user interfaces and an ERC Advanced Grant in 2024 for research on computational models of humans.

Andrew Howes is Professor of computer science at University of Exeter. He was previously FCAI Visiting Professor, Marshall M. Weinberg, Visiting Professor, Department of Psychology, University of Michigan in 2016 and visiting professor at NASA Ames Research Center in 2002 and 2015. Howes is a leader in computational rationality. Howes's interdisciplinary work has been published in the most prestigious journals in Cognitive Science and in HCI and 5 of his papers have won honorable mentions at ACM CHI. He has been associate editor at Cognitive Science journal and International Journal of Human-Computer Studies. He was papers chair of the 2017 33rd International Meeting of the Cognitive Science Society and is a former Head of Computer Science at the University of Birmingham.

4 Resources

Projects and other material by the instructors that take use of computational cognitive modeling. Some of these models will also be employed in the course. The main material for the course is accessible at <https://github.com/jussippjokinen/CogMod-Tutorial>, and other resources include:

- <https://github.com/aikkala/user-in-the-box/tree/main>
- <http://www.dcs.gla.ac.uk/~jhw/computationalinteraction/>
- <https://userinterfaces.aalto.fi/touchscreen-typing/>
- <https://users.aalto.fi/~jokinej10/visual-search/>
- <https://github.com/aditya02acharya/TypingAgent>
- <https://gitlab.com/jokinenj/multitasking-driving>

5 Accessibility

The course makes heavy use of Python notebooks, which fortunately offer a large range of accessibility tools. For instance, the Accessibility Toolbar (AT4N, <https://pypi.org/project/accessibility-toolbar/>) supports dyslexic and visually impaired users.

References

- [1] Xiuli Chen, Aditya Acharya, Antti Oulasvirta, and Andrew Howes. 2021. An adaptive model of gaze-based selection. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. 1–11.
- [2] Jussi PP Jokinen, Aditya Acharya, Mohammad Uzair, Xinhui Jiang, and Antti Oulasvirta. 2021. Touchscreen typing as optimal supervisory control. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. 1–14.
- [3] Jussi PP Jokinen, Tuomo Kujala, and Antti Oulasvirta. 2020. Multitasking in driving as optimal adaptation under uncertainty. *Human Factors* (2020), 18. <https://doi.org/10.1177/0018720820927687>
- [4] Antti Oulasvirta, Jussi PP Jokinen, and Andrew Howes. 2022. Computational Rationality as a Theory of Interaction. In *CHI Conference on Human Factors in Computing Systems*. 1–14.
- [5] Danqing Shi, Yujun Zhu, Jussi PP Jokinen, Aditya Acharya, Aini Putkonen, Shumin Zhai, and Antti Oulasvirta. 2024. CRTypist: Simulating Touchscreen Typing Behavior via Computational Rationality. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–17.